
Supplementary Material for “BAR: Budgeted Actor Refinement Decouples Bootstrap Exposure from Policy Reach”

A FROM LOCAL WASSERSTEIN FLOW TO FINITE-STEP TRAINING

A.1 Behavior anchor and Proposition 1

Let $b(s) = \mathbb{E}[a \mid s]$. The conditional bias–variance identity

$$\mathbb{E}[\|\mu(s) - a\|_2^2 \mid s] = \|\mu(s) - b(s)\|_2^2 + \mathbb{E}[\|a - b(s)\|_2^2 \mid s]$$

proves Eq. (7) after averaging over states and dividing by d ; its second term is independent of μ . For deterministic policies, $W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2$, so $\mathbf{d}_d^2$ is a fixed rescaling of integrated squared Wasserstein distance on conditional Dirac policy maps with the state marginal held fixed; it is not transport between full policy-induced occupancy measures. Dividing the normalized anchored objective $-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$ by α gives Eq. (8) with $h = \alpha/2$. More generally, $-\lambda \mathbb{E}_s \hat{Q}$ gives $h = \lambda C/2$ because $\hat{Q} = C\bar{Q}$.

A.2 Local explicit–implicit equivalence

Let $\beta = dh/C$, $g = \nabla_a \hat{Q}(a_0)$, and suppose the action constraint is inactive. The action-metric-matched linearized update is $\Delta a_{\text{lin}} = \beta g$. A statewise exact proximal solution satisfies

$$\Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (1)$$

If the Hessian of $\hat{Q}$ is locally Lipschitz near a_0 ,

$$\nabla_a \hat{Q}(a_0 + \Delta a) = g + H\Delta a + O(\|\Delta a\|^2). \quad (2)$$

Thus

$$(I - \beta H)\Delta a_{\text{prox}} = \beta g + O(\beta \|\Delta a_{\text{prox}}\|^2). \quad (3)$$

For $\beta \|H\| < 1$ and sufficiently small β ,

$$\Delta a_{\text{prox}} = \beta g + \beta^2 Hg + O(\beta^3), \quad (4)$$

whereas $\Delta a_{\text{lin}} = \beta g$. With only C^2 regularity, the safe remainder is $o(\beta^2)$ rather than $O(\beta^3)$. Both ideal updates are first-order schemes for fixed positive C , and changing a fixed C only rescales time. A positive policy-dependent global C preserves the raw-critic ascent direction under an adaptive time parameterization in the infinitesimal limit. This derivation is not a comparison of two maps with different finite-step normalizations, and is not applied to the large-budget regime with active projection, ReLU-region crossings, critic drift, or inaccurate actor solves.

A.3 Fixed-budget composition for an ideal consistent map

Proposition 1 (Local fixed-budget composition). *Let a smooth one-step update in local coordinates satisfy $\Psi_h(x) = x + hf(x) + h^2p(x) + O(h^3)$. For fixed K and $T \rightarrow 0$,*

$$\begin{aligned} \Psi_{T/K}^K(x_0) &= x_0 + Tf_0 \\ &\quad + T^2 \left[\frac{K-1}{2K} J_0 f_0 + \frac{1}{K} p_0 \right] + O(T^3), \end{aligned} \quad (5)$$

where $f_0 = f(x_0)$ and $J_0 = Df(x_0)$. Coordinate explicit Euler has $p_0 = 0$, whereas exact backward Euler has $p_0 = J_0 f_0$. Hence both are first-order methods with leading fixed- K flow bias proportional to $1/K$.

Proof. Write $x_j = x_0 + jhf_0 + O(h^2)$ for fixed K , which follows inductively from the one-step consistency assumption. Smoothness then gives

$$f(x_j) = f_0 + jhJ_0f_0 + O(h^2). \quad (6)$$

Summing the K increments,

$$\begin{aligned} x_K - x_0 &= h \sum_{j=0}^{K-1} f(x_j) + h^2 \sum_{j=0}^{K-1} p(x_j) + O(Kh^3) \\ &= Khf_0 + h^2 \frac{K(K-1)}{2} J_0f_0 \\ &\quad + Kh^2 p_0 + O(h^3). \end{aligned} \quad (7)$$

where the final remainder uses fixed K . Substituting $h = T/K$ proves Eq. (5). The exact flow satisfies $x(T) = x_0 + Tf_0 + \frac{1}{2}T^2J_0f_0 + O(T^3)$. Thus explicit Euler has leading error $-T^2J_0f_0/(2K)$, while backward Euler has $+T^2J_0f_0/(2K)$. Both remain first-order methods.

This proposition repeats one consistent near-identity map from its own output. It does not describe independently initialized persistent actors, whose parameter-gradient update can remain nonzero at $h = 0$.

The trained critic uses ReLU hidden units. At fixed state it is piecewise affine in action, so its action gradient is constant inside one activation region. Within such a region, the ideal explicit update and the same-region stationary backward-Euler branch coincide exactly; this does not assert that a global proximal minimizer cannot jump to another region. At the first activation boundary, the smooth-field assumptions used in the expansion above no longer apply. The learned-critic follow-up therefore audits activation-region residence and boundary incidence rather than fitting a nonzero $1/K$ error slope.

A.4 Inexact estimated-energy margin and critic-error transfer

By ε_k -optimality relative to the feasible comparator μ_{k-1} ,

$$\begin{aligned} &-\frac{2h_k}{C_k} \mathbb{E}_s \widehat{Q}(s, \mu_k(s)) + d_d^2(\mu_k, \mu_{k-1}) \\ &\leq -\frac{2h_k}{C_k} \mathbb{E}_s \widehat{Q}(s, \mu_{k-1}(s)) + \varepsilon_k. \end{aligned} \quad (8)$$

Rearranging yields Eq. (13). For $\widehat{Q} = Q^\dagger + e$, substitution further gives

$$\begin{aligned} m_k^{\widehat{Q}} &= \frac{2h_k}{C_k} \mathbb{E}_s [Q^\dagger(\mu_k) - Q^\dagger(\mu_{k-1})] - d_d^2(\mu_k, \mu_{k-1}) \\ &\quad + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})] \\ &= m_k^{Q^\dagger} + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})]. \end{aligned} \quad (9)$$

This proves Eq. (14). If $|e(s, a)| \leq \delta_k$ on the realized path, then

$$m_k^{Q^\dagger} \geq -\varepsilon_k - \frac{4h_k}{C_k} \delta_k. \quad (10)$$

The result concerns a trusted critic objective. A true-return statement would require additional occupancy and value-estimation assumptions.

A.5 Proof of Proposition 2

Let $F_{s'}(a) = \min_j Q_j^-(s', a)$. If every $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz, then

$$\begin{aligned} |F_{s'}(a) - F_{s'}(b)| &\leq \max_j |Q_j^-(s', a) - Q_j^-(s', b)| \\ &\leq L(s') \|a - b\|_2. \end{aligned} \quad (11)$$

Table 1: Verified one-stack actor-phase profile. Peak is the fresh-worker backend process-lifetime high-water mark, not an isolated actor-call peak.

K	Time (ms)	Time/ $K=1$	Peak (bytes)	Peak/ $K=1$
1	1.434567	1.000	9,684,736	1.000
2	1.542470	1.075	14,680,064	1.516
3	1.685672	1.175	15,627,520	1.614
4	1.887347	1.316	18,045,440	1.863

Therefore,

$$|y_{\mu_1} - y_{\mu_0}| \leq \gamma L(s') \|\mu_1(s') - \mu_0(s')\|_2. \quad (12)$$

Squaring, averaging over dataset transitions, and taking the square root proves Eq. (15). Later actors do not occur because the TD target conditions only on μ_1 .

B EXPERIMENTAL DETAILS

B.1 Algorithms and schedules

The implementation parameter is $T = \tau = \alpha/2$, and the normalized budget is split as $h = T/K$. Thus $K = 1$ recovers TD3+BC with coefficient $\alpha = 2T$. BAR-Prox uses Eq. (9) with a moving reference; BAR-Lin constructs Eq. (16) and uses MSE regression. The K actor networks are independently initialized persistent networks with separate Adam states. Each iteration first updates the online critic. On every second iteration, one optimizer step is then performed per actor in chain order on the same minibatch: actor one updates first; its target actor and the updated critic each receive a Polyak update of .005; later actors then use stop-gradient references from the updated online chain. Equations (5) and the main-text linearized scale define the actual normalization, with safeguard 10^{-6} . All methods share data, critic architecture, and 10^6 critic updates, while a K -hop run performs K actor optimizer steps and is not compute matched.

Measured actor-phase cost. A locked profile held one NVIDIA H200, JAX/JAXlib 0.10.2, `hopper-medium-v2`, batch size 256, $T = 12$, initialization, and the sampled batch fixed while varying only K . Table 1 reports the median synchronized dispatch of the compiled full K -actor production phase. It excludes lowering, compilation, warmup, and critic optimization.

The peak covers dataset/batch transfer, actor/critic-state initialization, compilation, warmup, and timed trials. Thus the measured $K = 4$ actor phase is 1.316 times $K = 1$ on this stack, while the scoped high-water mark is 1.863 times $K = 1$. The raw archive and independent verifier are released. This profile quantifies practical overhead for the tested stack; it is not an end-to-end training-time or hardware-independent memory law.

The ideal proximal map is not a certificate for this live chain. For a later persistent actor that differs from its predecessor, even $h = 0$ leaves

$$\nabla_{\theta_k} \mathbb{E}_s [\|\mu_k(s) - \text{sg } \mu_{k-1}(s)\|_2^2/d] \neq 0 \quad (13)$$

in general. One Adam step therefore need not become a near-identity map as the nominal coefficient vanishes; the ideal analysis supplies local geometry, while the sweep evaluates the implemented optimizer trajectory.

We use a 256–256 ReLU actor MLP with tanh output and twin 256–256 ReLU critic MLPs, batch size 256, Adam with learning rate 3×10^{-4} , discount .99, policy delay 2, target noise .2 and clip .5 times the action bound, and state normalization by dataset mean and standard deviation plus 10^{-3} . Every reported sweep score is the ten-episode D4RL normalized mean at the final 10^6 -update checkpoint. Archived configurations record 50,000-update intermediate evaluation for K1/P2/P3, mixed 50,000/final-only evaluation for L2, and final-only evaluation for L3. Exact historical per-run configurations for proximal seeds 2–3 were not retained; post-hoc diagnostic manifests record audit settings, not the training launch. Intermediate cadence does not enter the headline estimand. The 14-value grid spans .05 to 20. Both proximal and linearized procedures use seeds 0–3 in the headline score grids.

The nine exact datasets are the Cartesian product of domains `halfcheetah`, `hopper`, and `walker2d` with suffixes `medium-v2`, `medium-replay-v2`, and `expert-v2`. Timeout rows are omitted from transition construction;

Table 2: Evidence map. Roles distinguish the complete headline score grids from post-hoc diagnostics and exploratory controls.

Evidence	Unit / scope	Seeds	Role
Proximal and linearized scores	9 tasks $\times$ 14 budgets	0–3	Headline grids
P3 target/final displacement	270 checkpoints; 90 matched contrasts	0–1	Post-hoc full audit
P4 target branch	180 checkpoints; 90 contrasts	0–3 / 0–1	Four-seed target archive
P3/P4 value + geometry	270 checkpoints; 90 per method	0–1	Post-hoc full audit
P3 two-actor control	90 pairs; 180 final scores	0–1	Exploratory mechanism control
P4 two-actor scores	90 pairs; two runtime stacks	0–1	Descriptive; inadmissible
ReLU residence	6 tasks; 6144 anchors	0–1	Prelocked scope audit
Route and component controls	8 selected runs	0–1	Post-hoc targeted controls
Actor cost	$K = 1\text{--}4$; H200	init. 0	Systems profile

terminal flags stop bootstrapping.

B.2 Diagnostic quantities

On a fixed current-state validation batch with paired dataset actions, we compute the empirical action-averaged proxies

$$\begin{aligned}\hat{D}_{\text{critic}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_1(s) - a\|_2^2, \\ \hat{D}_{\text{final}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_K(s) - a\|_2^2.\end{aligned}$$

These quantities measure the online actor chain relative to its sample anchor. They are proxies for, rather than direct measurements of, the next-state target-policy displacement in Proposition 2: the practical TD target uses a Polyak target actor and policy-smoothing noise. By Eq. (7), the population expectation of each proxy equals squared displacement from the conditional-mean behavior map plus the same additive conditional-action variance. Population differences therefore cancel this common floor, but ratios do not: a ratio of sample-anchored proxies is not a ratio of mean-policy displacements. Finite validation batches add sampling error to both interpretations. We additionally evaluate the deterministic Polyak target actor and 64 draws of clipped target-smoothing noise on 4096 next-state transitions per checkpoint. For source index i , the audit pairs $s'_i = s_{i+1}$ with dataset action a_{i+1} ; it excludes timeout rows and excludes true terminal rows from this bootstrap-only sample. All target-exposure quantities remain squared displacement per action coordinate relative to the paired next dataset action. Thus absolute levels include conditional action variance, whereas paired differences use a common sampled anchor. For the operational value-space readout, a second protocol pairs all 90 TD3+BC, P3, and P4 cells on the same next-state batch and clipped smoothing noise and records

$$\Delta_y^{\text{RMS}} = \gamma \sqrt{\mathbb{E} \left[\left(\min_j Q_j^-(s', \tilde{\mu}_1(s')) - \min_j Q_j^-(s', \tilde{a}'_{\mathcal{D}}) \right)^2 \right]}.$$

The within-run target critic is the operational training object. A matched TD3+BC ($K = 1$) target critic from the same task, T , and seed is also applied to all three actors to isolate the action-side contrast under one learned scorer. Neither critic is ground truth. For BAR-Lin, projection attenuation is the RMS norm ratio between the clipped and raw target displacements. If $\tilde{g}$ is the normalized critic action gradient and Δa is the learned actor displacement, define

$$h_{\text{eff}} := \frac{\mathbb{E} \langle \Delta a, \tilde{g} \rangle}{\mathbb{E} \|\tilde{g}\|_2^2}.$$

We report $h_{\text{eff}}/(dT/K)$ as the effective fraction of the nominal linear-target coefficient. The matched audit evaluates 512 fixed states per environment–seed run. We also record action saturation, TD-error quantiles, critic magnitude, critic loss, and gains under both the jointly trained critic and a controlled common critic: the frozen final TD3+BC $T = 1$ critic from a separate canonical reference run outside the fourteen-value sweep, matched by environment and seed. A run-total gain is the sum of mean Q changes across the K hops on this shared batch. The common critic supports cross-critic comparison but is not treated as ground truth. A run is descriptively labeled collapsed only from its external normalized return, using the threshold 20.

For the exploratory simulator-reference audit, we clone MuJoCo integration state and environment RNG from evaluation-policy trajectories, record the episode clock, and reset the TimeLimit for a continuing-task critic-calibration estimand. At each state we compare $a_1 = \mu_1(s)$ and $a_K = \mu_K(s)$ under continuations ρ_1 , μ_1 , and μ_K , where ρ_1 is the Polyak first target actor with TD3 clipped target-smoothing noise. The primary calibration target is the discounted return under this effective backup continuation. The screening covers three environments, $T \in \{4, 7, 10, 14, 20\}$, $K \in \{2, 3\}$, two seeds, and 12 states per checkpoint. It uses a 1000-step horizon, $\gamma = .99$, four common-random-number target-smoothing rollouts, and a one-return-unit ranking dead zone. It is a post-hoc screening study, not a causal routing intervention.

The route-only intervention trains two actor-detached shadow critics from identical initial parameters on the same minibatches and target-smoothing draws while the primary first-hop critic continues to drive the shared actor trajectory. The four environment–budget cells—Hopper-medium at $T \in \{10, 20\}$ and Walker2d-medium at $T \in \{4, 7\}$, each with seeds 0–1—were selected post hoc from visible stability-frontier transitions, so inference is descriptive of this targeted sample. One shadow uses the first-hop target actor and the other the final-hop target actor. The simulator audit evaluates 12 cloned states at four checkpoints in each of eight runs; inference aggregates checkpoint-level state medians to the training-run level.

For route-native evaluation, ρ_1 and ρ_K are respectively the Polyak first and final target actors with the same TD3 clipped target-smoothing distribution. The common-continuation rerun instead evaluates both shadow critics under ρ_1 on the same sampled states and rollout seeds.

The compute-matched study uses the same eight final cells. Sequential $K = 3$ performs actor updates anchored at μ_0, μ_1 , and μ_2 ; the fixed-reference control anchors both later actors at μ_1 ; the direct-final control applies two optimizer calls to one $2T/3$ objective anchored at μ_1 . Every condition performs three actor optimizer calls per delayed update. The target-lag control instead keeps $K = 1$ and changes only the actor-target Polyak rate from .005 to .005/3, leaving the critic-target rate unchanged.

The broader policy-separation control covers all nine tasks, $T \in \{4, 7, 10, 14, 20\}$, and seeds 0–1 at one million updates. Its independently initialized target and evaluation actors are both anchored directly to dataset actions. The target actor uses coefficient $T/3$ and alone supplies Bellman targets; the deployed actor uses T . We pair it by task, budget, and seed with a contemporaneous BAR-P3 rerun under the same evaluation protocol. It performs two rather than three actor optimizer calls and follows a separately realized critic trajectory. It is therefore an MCEP-inspired policy-separation mechanism control, not a published-MCEP reproduction, a compute-matched re-centering ablation, or an exact shared-trajectory comparison.

The ReLU residence audit freezes the final $T = 1$ Q1 critics for the six Hopper and Walker2d tasks and seeds 0–1. For each run it deterministically selects 512 previously unused dataset-action anchors satisfying $\max_j |a_j| \leq .95$, after excluding every archived and development index. Along the normalized critic-gradient ray it solves the first ReLU activation or action-box intersection exactly. The sampled rows are fresh, but the critics and task families were studied previously; this is a prelocked fresh-index scope audit rather than a checkpoint-level holdout.

B.3 Estimands and uncertainty

The primary high-budget estimand uses the upper seven values of the ordered fourteen-point grid, $T \geq 4$. Within each task it averages the available seeds and those seven budgets, then weights the nine task means equally. This is an operational stress-region aggregate, not a claim of pointwise monotonicity in T or task. A seed-mean environment–budget cell is called collapsed when its normalized score is below 20. Thresholds 0, 10, 20, 30, and 40 expose sensitivity to that descriptive boundary.

For score contrasts we resample the nine task-level high-budget means with replacement for 100,000 draws using analysis seed 20260830. Raw-run collapse risk is first averaged within each task and then resampled by task using the same draw count and seed. These 95% percentile intervals preserve the complete within-task budget and seed structure but do not remove dependence among the three datasets sharing one dynamics family. With only three families, family analyses are sensitivity checks rather than a three-cluster bootstrap. A secondary task-then-seed bootstrap uses 200,000 draws and seed 20260902: after averaging the fixed budget set within each seed, it resamples nine tasks and independently resamples four seeds with replacement for each procedure and each task occurrence. This targets marginal procedure means rather than treating seed labels as exact trajectory pairs. Cross- K means estimate each randomized procedure’s marginal performance. Identical critic initialization across depth is not required for validity; common random numbers would reduce variance. Paired rescue counts

Table 3: Task-level full-horizon ReLU-region residence. Each entry pools two seeds (1024 eligible interior anchors).

Task	.025	.05	.1	.2
Hopper medium	.985	.970	.950	.912
Hopper replay	.946	.890	.793	.641
Hopper expert	.997	.993	.990	.980
Walker medium	.958	.924	.857	.751
Walker replay	.850	.713	.516	.310
Walker expert	.987	.975	.943	.896
Task-equal	.954	.911	.842	.748

Table 4: Mean score over nine tasks at each total budget. Every row uses four seeds.

T	.05	.1	.2	.4	.7	1.5	2.5	4	7	10	12	14	17	20
TD3+BC	63.81	62.57	64.58	65.94	67.69	74.79	67.41	45.93	30.36	23.77	21.64	18.96	18.92	18.09
Prox-2	60.45	64.19	67.90	69.61	73.65	78.05	80.66	74.48	58.88	41.14	32.17	29.55	25.95	26.25
Prox-3	60.14	60.82	68.08	71.01	73.32	78.95	81.41	81.26	71.00	51.14	49.74	41.68	32.75	35.83
Prox-4	59.59	63.95	68.71	71.26	73.26	77.70	81.25	82.06	81.79	74.35	67.88	54.28	52.14	35.29
Lin-2	62.28	63.45	66.62	71.63	70.98	75.13	78.43	75.16	51.49	37.81	26.03	27.46	27.50	23.55
Lin-3	59.27	62.86	64.99	70.90	74.13	77.74	78.20	78.16	67.80	45.60	39.96	41.69	26.28	25.59

remain descriptive under the realized seed coupling.

C REGIME-BASED ADDITIONAL RESULTS

C.1 Small-horizon flow scope on learned critics

The JKO/gradient-flow reading is local: it requires a fixed critic, a common frozen scale for the maps compared, an exact statewise update, inactive projection, and no activation-boundary crossing. The T^2/K composition statement additionally repeats one consistent near-identity map. The following prelocked audit asks how often its sharpest learned-critic premise, residence in one ReLU region, holds along a critic-gradient ray. It does not use return to define success.

The clean-source snapshot verifier independently reconstructs all 6144 activation geometries and 27,648 finite-difference coordinates. At the longest probe horizon, .2, 1544 anchors first cross a ReLU boundary, three cross the action box, and none is nonfinite or boundary-ambiguous. The high small-horizon residence makes the exact local affine identity a relevant description, while the .310–.980 task range at the .2 horizon already warns against uniform extrapolation. The live training sweep below tests what happens after this local description ceases to organize the outcome.

C.2 Return phase diagram: plateau, transition, and large-budget tail

The score grid gives a concrete regime boundary. At $T = .05, .1, .2$, the six procedures occupy a narrow band and their ordering changes with T ; this is compatible with a shared local direction, not evidence that deeper discretization improves return. TD3+BC reaches its maximum at $T = 1.5$, then drops from 67.41 at $T = 2.5$ to 45.93 and 30.36 at $T = 4$ and 7. P4 instead gives 81.25, 82.06, 81.79 on those three budgets. We use this $T = 2.5$ –7 region descriptively as the observed transition. For $T \geq 10$, all methods degrade and the setting is plainly nonlocal; P4 remains strongest through $T = 17$, but P3 and P4 are nearly tied at $T = 20$. Thus depth delays the failure frontier rather than dominating at every budget. The residence audit stops at probe horizon .2, while the direct projection diagnostic is at $T = 20$; they bracket rather than identify the mechanism of the observed $T = 2.5$ –7 transition.

Practical budget summaries. At TD3+BC’s best shared grid budget, $T = 1.5$, P3 and P4 score 78.95 and 77.70 versus 74.79; the descriptive method-wise grid maxima are 82.06 for P4 at $T = 4$ and 74.79 for TD3+BC at $T = 1.5$. To reduce same-grid selection optimism, a post-hoc leave-one-dynamics-family-out sensitivity selects

Table 5: Collapse-count sensitivity for $T \geq 4$. Entries are seed-mean cells / raw seeded runs, with denominators 63 / 252 for every method.

Threshold	TD3	Prox-2	Prox-3	Prox-4	Lin-2	Lin-3
0	1/20	0/14	0/13	0/10	1/17	2/23
10	30/134	22/103	15/81	6/50	24/105	17/85
20	34/142	22/111	16/84	8/54	25/112	21/95
30	36/149	26/115	18/88	11/55	29/120	24/99
40	38/165	30/120	19/96	14/58	32/125	24/105

Table 6: Per-environment high-budget ($T \geq 4$) means.

Environment	TD3	Prox-2	Prox-3	Prox-4	Lin-2	Lin-3
Hopper-M	11.1	28.7	63.7	84.7	28.0	54.0
Hopper-MR	45.6	92.3	99.6	99.6	93.0	99.0
Hopper-E	3.1	16.8	28.2	46.4	20.8	27.3
HalfCheetah-M	58.8	58.9	58.9	58.2	53.2	55.8
HalfCheetah-MR	51.0	51.8	51.8	52.6	46.3	48.8
HalfCheetah-E	19.9	23.3	30.5	32.8	20.6	27.6
Walker2d-M	9.3	26.0	37.2	59.8	9.0	24.9
Walker2d-MR	27.3	51.2	72.4	76.8	53.4	52.3
Walker2d-E	2.2	21.8	25.1	65.1	21.6	28.2

one T per method on the other six tasks and evaluates it on the three held-out tasks. TD3+BC selects $T = 1.5$ in all folds; P4 selects $T = 2.5, 7, 4$ when Hopper, HalfCheetah, and Walker2d are held out. The corresponding P4 minus TD3 family differences are +17.08, -8.94 , +2.83, giving 78.44 versus 74.79 overall (+3.65; 7/9 task differences positive). This transfer sensitivity is not a pre-registered selection rule and exposes material family dependence.

A task-then-independent-seed bootstrap sensitivity preserves the high-budget P4-P3 and L3-L2 conclusions, with intervals [3.04, 22.73] and [2.90, 14.07]. At the common $T = 1.5$, P4 minus TD3+BC is +2.92 with interval $[-.20, 8.46]$, so that fixed-budget practical contrast remains unresolved at four seeds even though its point estimate is positive.

Beyond the main-text P4-P3 contrast, P4 minus TD3+BC gains 38.59 points with task-resampling interval [21.39, 54.63] and eight of nine positive task means. At the score-20 threshold, the P4-P3 and L3-L2 raw collapse-risk differences are $-.119$ ($[-.218, -.024]$) and $-.067$ ($[-.131, -.004]$). BAR-Lin is positive in 45/63 continuous cell contrasts (median +1.37); its five rescues and one reversal account for 42.8% of the total gain. Thresholded transitions remain descriptive.

C.3 What changes at the transition: lower-tail rescue and critic validity

The return separation could come from a uniform score lift or from preventing specific unstable trajectories. Collapse transitions and diagnostic medians favor the second explanation for proximal P4: its cellwise median gain over P3 is small, while a few lower-tail rescues account for most of the aggregate improvement. The external score defines failure; critic quantities below are post-hoc signatures of the learned trajectory.

A restricted four-environment seed-2/3 archive (80 P2/P3 runs) reproduces the reference-critic split: all 45 stable gains are positive and all 35 collapsed gains negative (medians +1.19 and -4.20). It is a sensitivity check of the same implementation, not an independent replication.

Both proximal seed pairs reproduce the strict depth ordering in Table 8. The linearized transition also repeats, with gains of 7.23 ([1.99, 12.47]) on seeds 0-1 and 8.80 ([4.01, 15.21]) on seeds 2-3. These are seed replications of one implementation.

Table 7: Core matched $K = 2$ diagnostic medians over $T \in \{4, 7, 10, 14, 20\}$. Stable and collapsed groups are defined by external return, while the remaining rows diagnose the learned trajectory.

	Lin st.	Lin col.	Prox st.	Prox col.
No. runs	52	38	59	31
TD p99	270	9.6×10^{23}	160.8	5.0×10^{21}
$\hat{D}_{\text{critic}}$	.112	.538	.064	.446
$\hat{D}_{\text{final}}$	.147	.626	.094	.578
Common- Q gain	+2.11	-1.42	+2.51	-1.06

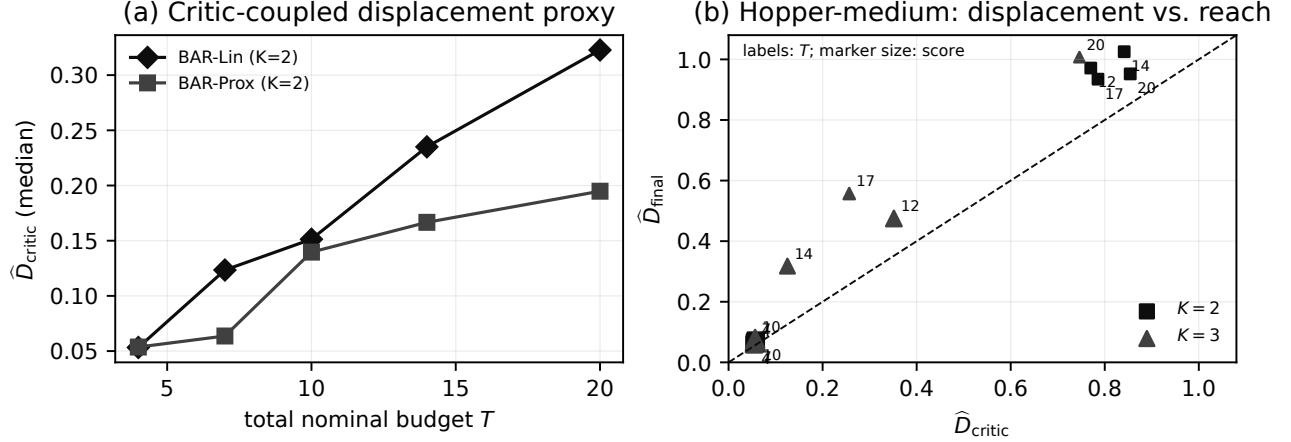


Figure 1: Realized movement. (a) The proximal realization produces a smaller current-state first-hop displacement proxy for $T \geq 7$. (b) On Hopper-medium, $K = 3$ either increases final reach at approximately fixed displacement or contracts both displacements in the observed score-transition region. The displayed current-state quantities exclude Polyak target lag and smoothing noise. Labels indicate T and marker size is proportional to score.

C.4 Which part of multi-hop matters?

BAR changes both how far the target-routed first branch moves and how a final policy is constructed. The controls below separate some alternatives, but no single one isolates every element of the persistent chain.

Across the 90 matched task-budget-seed cells, the two-actor control scores 59.22 versus 57.67 for the contemporaneous BAR-P3 rerun; raw collapse counts are 21/90 and 23/90, and both have 9/45 seed-mean collapsed cells. The control-minus-BAR task-equal mean is +1.56, with a 95% task-resampling interval $[-1.52, 5.44]$; only 2/9 task means favor the control. At the run level the control wins 34 cells, BAR wins 52, and four tie. The positive mean but negative paired median reflects tail redistribution rather than a broad cellwise shift: the mean differences are +5.53 for seed 0 and -2.42 for seed 1, and the two seeds have opposite signs in 30/45 task-budget pairs. Using normalized return below 20 as collapse, the control rescues five BAR collapses and introduces three at the run level. After averaging the two seeds first, both procedures collapse in 9/45 cells, and none of the 45 paired classifications changes.

Eight Walker2d-expert cells resumed from the matching logged emergency-checkpoint steps on the CPU recovery path, and the two $T = 20$ cells began there from zero. Removing the entire ten-cell task changes the mean difference to +1.83; because this also removes a task, it is not a hardware-effect estimate. The contemporaneous BAR-P3 aggregate is only .14 above the released P3 aggregate on these cells, but their cellwise differences have RMSE 24.03 and range from -99.50 to +98.63. We therefore pair the control only with the contemporaneous rerun. The higher observed control mean and identical seed-mean collapse count show that sequential re-centering is not a prerequisite for those two aggregate outcomes on this grid. The bundled comparison establishes neither equivalence nor which simpler-procedure element is responsible.

The matched P4 score experiment also completed 90 pairs across two hosts. The score-level BAR-minus-control

Table 8: Host-separated full-grid replication. High is the $T \geq 4$ mean; Cell is the number of seed-pair-mean cells below 20 out of 63.

K	High 0-1	High 2-3	Cell 0-1	Cell 2-3
1	25.97	24.79	35	35
2	42.02	40.38	25	26
3	52.78	51.05	17	16
4	63.61	64.33	9	10

Table 9: Targeted final-return controls on the same eight environment-budget-seed cells. The three $K = 3$ rows are compute matched to three actor optimizer calls; the $K = 1$ rows differ only in actor-target Polyak rate.

Condition	Mean	Median	Score < 20
Sequential $K = 3$	71.4	88.9	2/8
Fixed-ref. $K = 3$	70.0	87.9	2/8
Direct $K = 3$	59.4	89.6	3/8
Standard $K = 1$	13.6	0.8	6/8
$3\times$ target lag, $K = 1$	12.0	1.5	7/8

task-equal mean is -2.31 , its paired median is $+.68$, BAR wins 55 cells versus 34 control wins and one tie, and the task-resampling interval is $[-7.25, 2.41]$, giving the predeclared numeric label *unresolved*. We do not promote this as the locked P4 result: the two shards use materially different resolved Python/JAX/Flax/Optax stacks, five retained Walker2d-expert runs have nonfinite optimizer states, and the tail raw checkpoints are not in the release host. Keeping the finite final scores avoids post-hoc outcome filtering; the merge remains a descriptive sensitivity.

The paired route-only study uses the same eight runs and aggregates four checkpoints per run. For critic route $r \in \{\text{first}, \text{final}\}$ and evaluation continuation ρ , define the checkpoint state-median symmetric relative error

$$E_r^\rho = \text{median}_s \frac{|Q_r(s, a_K) - Q_{\text{MC}}^\rho(s, a_K)|}{|Q_r(s, a_K)| + |Q_{\text{MC}}^\rho(s, a_K)| + 10^{-12}}. \quad (14)$$

The route-native run statistic averages $E_{\text{final}}^{\rho_K} - E_{\text{first}}^{\rho_1}$ over four checkpoints: final-route error is larger in all four-checkpoint aggregates but only five of eight final checkpoints. The run-level mean is .124, median .013, exact two-sided sign-flip $p = .0078$, and run-bootstrap mean 95% interval $[.008, .323]$. Because this native comparison changes both critic route and continuation, it is not a common-value comparison. We therefore reran both critics under ρ_1 . The common-continuation difference is positive in six of eight runs, with mean .116, median .0022, sign-flip $p = .164$, and bootstrap interval $[.0003, .317]$. The pre-specified support rule requires a positive mean and two-sided exact sign-flip $p < .05$; the common-continuation result fails the second condition. Neither comparison estimates a return effect because the shadows do not feed back into actor training.

C.5 Direct exposure, reach, and critic-space response

The original direct audit covers 270 method checkpoints: TD3+BC, P2, and P3 over nine environments, five budgets, and seeds 0-1. A four-seed dump covers all 180 P4 target branches. Protocol v2 additionally evaluates same-next-state target and final geometry for 90 matched TD3+BC, P3, and P4 checkpoints. Table 10 reports this latter audit. Ratios are descriptive ratios of sample-anchored squared displacement, not conditional-mean policy distances. The final-to-target ratios below are separately reported on the RMS scale.

The deployed actor lies farther from the paired dataset action than the Polyak first actor in all 90 P3 and all 90 P4 cells; median final-to-target RMS ratios are 1.297 and 1.365. Under the common TD3+BC target critic with identical smoothing noise, Δ_y^{RMS} is lower than TD3+BC in 86/90 P3 cells (median ratio .717) and 88/90 P4 cells (.617). Every task median is below one; removing smoothing preserves the counts and gives ratios .702 and .607.

Absolute task-equal RMS differences are not used as typical-effect summaries. Within-run TD3+BC critics have mean $|Q(s', a'_D)| > 10^9$ in 31/90 cells, all externally collapsed, and log critic scale correlates .999 with log Δ_y^{RMS} .

Table 10: Matched-cell audit of deterministic next-state Polyak-target and final-policy displacement. The Lower column counts runs in which the left method has the smaller quantity.

Comparison	Quantity	Lower	Ratio
P3 / TD3	Target disp.	88/90	.650
P3 / TD3	Final disp.	37/90	1.060
P4 / TD3	Target disp.	89/90	.565
P4 / TD3	Final disp.	44/90	1.025

These finite but extreme tails dominate arithmetic means. The common-critic paired ratios therefore support an action-side critic-space perturbation claim, not critic accuracy or return causality.

The sampled feasible-comparator premise is also mixed: before one post-hoc Adam step, 85/180 P3 and 64/270 P4 later-hop rows need no positive slack; the remaining median slacks are 5.19×10^{-4} and 1.46×10^{-3} and decrease in 95/95 and 205/206 positive-slack rows. Because this does not replay the historical minibatch or live update, we do not interpret it as an ε_k bound or implementation certificate.

A restricted four-environment seed-2/3 extension reproduces the P3/P2 target-action-displacement direction in 36/40 cells (median ratio .905). It is a post-hoc sensitivity check, not a new full-grid estimand.

Across all 180 four-seed P4 target-branch cells, target-displacement medians are .051 for 144 stable runs and .130 for 36 collapsed runs. The full seed-0/1 final geometry shows no typical final contraction: its P4/TD3 median ratio is 1.025. For P3 the corresponding median is 1.060; a few large contractions can still make arithmetic mean differences negative.

The current-state proxy and deterministic target exposure have Pearson correlation .9999 over all 270 cells; their median ratio is .997 and the largest cellwise relative difference is 5.7%. The corresponding correlation after target smoothing is .9993. The close alignment is an empirical property of these checkpoints, where Polyak lag is small, rather than a mathematical equivalence. Among the 84 Prox-3/Prox-2 runs in which Prox-3 lowers deterministic target exposure, Prox-3 is nevertheless worse by more than one score point in 17. This rejects the stronger interpretation that minimizing exposure alone determines performance.

Across 55 stable and five collapsed checkpoints (720 states), the median checkpoint-level signed critic errors are -41.4 and $+1.70 \times 10^{12}$; 58/60 episode-cluster intervals exclude zero. Restoring episode clocks changes 21 checkpoint medians but preserves all 60 signs (overall median -36.4). The action-ranking subset is too sparse for a population claim: only 69/720 states clear the dead zone for both actions, with 29 sign disagreements.